Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 38)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

0 1 . 2 Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

0 1 . 3 Explain **three** ways in which coral reefs are important to humans.

[3 marks]

1

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0 2 . 1 Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed .....

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Sea lice and disease .....

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Escapes affecting wild gene pools .....

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Organic waste pollution .....

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0 2 . 2 Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

[2 marks]

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0 3 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 3 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 3 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

1 .....

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2 .....

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5

0 4 Table 1 lists some techniques used in ecological monitoring.

0 4 . 1 Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

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0 4 . 2 Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

0 4 . 3 Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

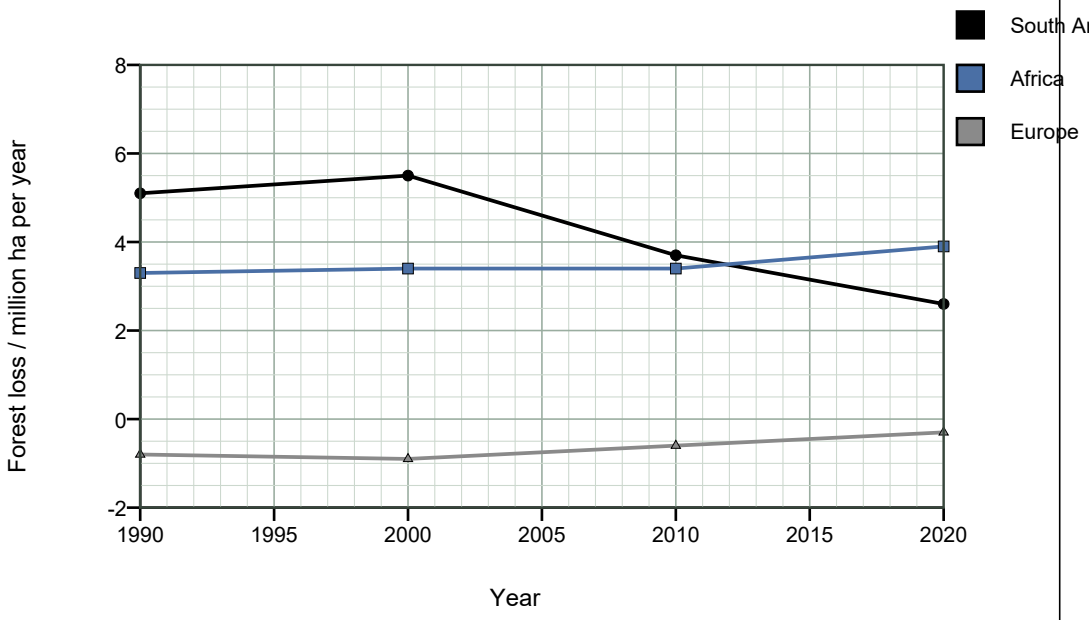
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0 5 Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 5 . 1 Describe the trends shown in Figure 1.

[3 marks]

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0 5 . 2 Explain the main causes of forest loss in South America and Africa.

[4 marks]

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0 5 . 3 Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

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0 5 . 4 Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

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0 6 . 1 Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

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0 6 . 2 Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

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2 .....

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4 .....

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0 6 . 3 Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

Blue whale .....

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Large blue butterfly .....

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10

0 7 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 7 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 7 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

5

0 8 . 1 Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0 8 . 2 Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

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0 8 . 3 Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

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0 9 . 1 Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

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0 9 . 2 Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

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0 9 . 3 Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

1 .....

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1 0 Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

1 0 . 1 Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

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1 0 . 2 Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

1 0 . 3 Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

1 0 . 4 Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

1 1 . 2 Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS